

=== CONFIGURACION GNB (1780312440) ===

```
# gNB Configuration for O-RAN E2E V2X
# This configuration uses srsRAN_Project for a 5G gNB.

# Core Network (AMF) connection
amf:
  amf_addr: 10.53.1.2 # IP address of the AMF (Open5GS 5GC)
  amf_port: 38412

# gNB Identity
mcc: 315
mnc: 010
tac: 7 # Tracking Area Code
nci: 0x000000010 # NR Cell Identity (e.g., 256 for cell ID 1)
gnb_id: 0x00001 # gNB ID

# Slice Support List (S-NSSAI)
tai_slice_support_list:
  - s_nssai:
      sst: 2 # Slice/Service Type for V2X (3GPP TS 23.501, Section 5.15.3.2)

# O-RAN CU-CP Configuration
cu_cp:
  bind_addr: 10.53.1.3 # IP address for CU-CP interface, must match Docker IP

# ZMQ Interface for Radio Link (for virtual RF)
zmq:
  tx_port: tcp://*:2000 # Port for transmitting to UE
  rx_port: tcp://0.0.0.0:2001 # Port for receiving from UE

# O-RAN DU-RF Configuration
du_rf:
  device_name: zmq # Use ZMQ for virtual radio interface
  device_args: "" # No specific device arguments for ZMQ
  srates: 23.04e6 # Sample rate (Hz), must match UE
  nof_antennas: 1 # Number of antennas
  tx_gain: 70 # Transmit gain (dB)
  rx_gain: 30 # Receive gain (dB)
  bands: [3] # NR band, must match UE
  channel_bandwidth_MHz: 20 # Channel bandwidth (MHz)

# Logging configuration
log:
  all_level: info
  filename: /tmp/gnb.log
  file_max_size: -1

# PCAP capture
pcap:
  enable: none
  mac_filename: /tmp/gnb_mac.pcap
  mac_nr_filename: /tmp/gnb_mac_nr.pcap
  nas_filename: /tmp/gnb_nas.pcap
```

=== CONFIGURACION UE (1780312440) ===

```
[rf]
freq_offset = 0
tx_gain = 80
rx_gain = 10
srate = 23.04e6
nof_antennas = 1
device_name = zmq
device_args = tx_port=tcp://*:2001,rx_port=tcp://localhost:2000,id=ue,base_srate=23.04e6
```

```
[rat.nr]
bands = 3
nof_carriers = 1
max_nof_prb = 106
nof_prb = 106
```

```
[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap
```

```
[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1
```

```
[usim]
mode = soft
algo = milenage
opc = 63BFA50EE6523365FF14C1F45F88737D
k = 00112233445566778899aabbccddeeff
imsi = 315010000000001
imei = 353490069873319
```

```
[rrc]
release = 15
ue_category = 4
```

```
[nas]
apn = srsapn
apn_protocol = ipv4
```

```
[gw]
```

```
[gui]
enable = true
```

=== DOCKER COMPOSE (1780312440) ===

```
version: '3.8'
networks:
  ran:
    ipam:
      config:
        - subnet: 10.53.1.0/24
services:
  5gc:
    build:
      context: open5gs
      target: open5gs
    container_name: open5gs_5gc
    privileged: true
    command: 5gc -c open5gs-5gc.yml
    volumes:
      - ./open5gs-5gc.yml:/open5gs/install/etc/open5gs/open5gs-5gc.yml
    networks:
      ran:
        ipv4_address: 10.53.1.2
  gnb:
    build:
      context: srsran_project
      target: srsran_project
    container_name: srsran_gnb
    privileged: true
    command: srsgnb -c gnb.yml
    volumes:
      - ./gnb.yml:/srsran_project/build/srsgnb/gnb.yml
    networks:
      ran:
        ipv4_address: 10.53.1.3
```